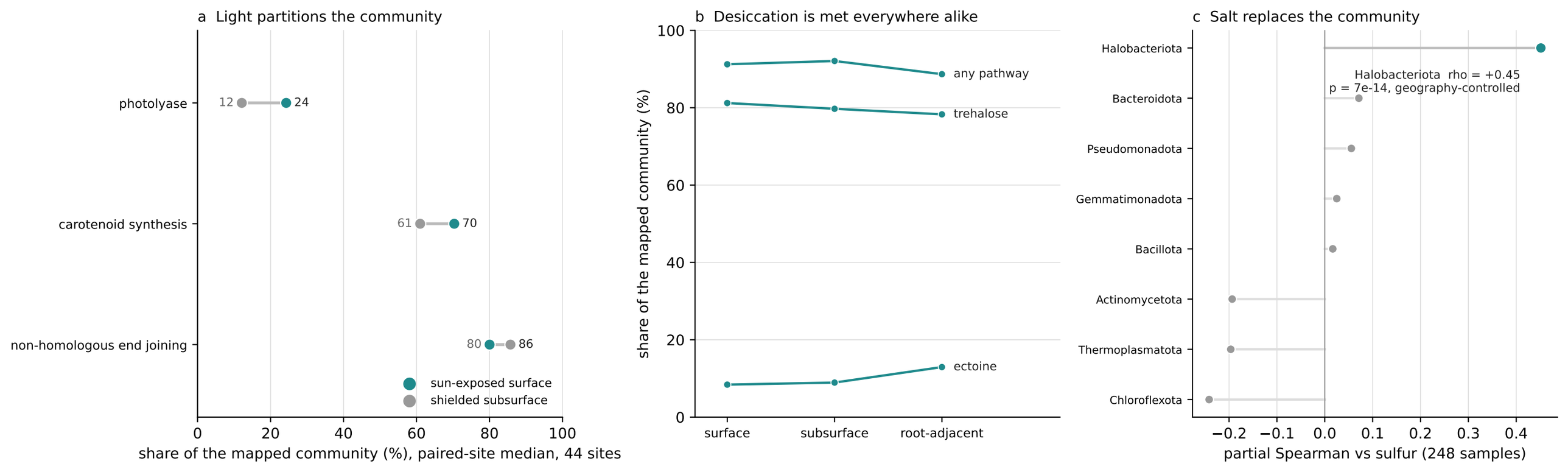
a Light partitions the community
photolyase
carotenoid synthesis
non-homologous end joining
12
24
61
70
80
86
sun-exposed surface
shielded subsurface
0
20
40
60
80
100
share of the mapped community (%), paired-site median, 44 sites
b Desiccation is met everywhere alike
share of the mapped community (%)
any pathway
trehalose
ectoine
surface
subsurface
root-adjacent
c Salt replaces the community
Halobacteriota
Bacteroidota
Pseudomonadota
Gemmatimonadota
Bacillota
Actinomycetota
Thermoplasmatota
Chloroflexota
Halobacteriota rho = +0.45
p = 7e-14, geography-controlled
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
partial Spearman vs sulfur (248 samples)